Lesson 7.1 Guided Notes

Session 1 · Integers and Opposites

I can write an integer for a real situation and name its opposite. · 6.NS.C.5

Word / Spanish	Meaning — complete the blank	Example
integer / entero	Zero, a counting number, or the _____ of a counting number.	−9, −1, 0, 6, 230
opposite / opuesto	A number the same _____ from 0, on the other side of 0.	−6 and 6
negative number / número negativo	A number _____ than 0, written with a − sign.	−9 is read “negative nine”
positive number / número positivo	A number _____ than 0.	6, 19, 230

Key idea

First decide what _____ means in the situation.

Gains, above, and deposits are _____. Losses, below, and debts are _____.

Worked example

In a game, 0 is your starting score. You earn 8 points, then you lose 8 points. Write each as an integer.

Zero means the _____ score.

Earning 8 points is _____. Losing 8 points is _____.

Each is 8 units from 0, on opposite sides.

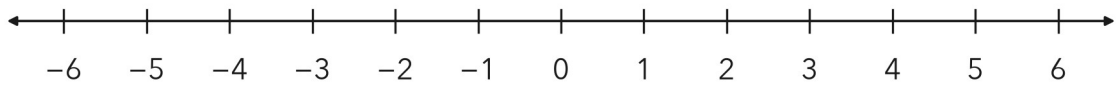
Answer: 8 and −8 are _____.

You try

1. Write an integer for each situation.
a. 14 feet below sea level _____
b. a gain of 6 pounds _____

_____

2. Plot −4 and its opposite. Label each point.



_____

Sum it up: To write an integer for a situation, I first ___, then I ___.

_____

Session 2 · Integers in Real Situations

I can interpret integers in real situations and find the opposite of an opposite. · 6.NS.C.5

Word / Spanish	Meaning — complete the blank	Example
opposite / opuesto	A number the same distance from 0, on the _____ side of 0.	−6 and 6
opposite of the opposite / opuesto del opuesto	Flip across 0 _____ and you are back where you started.	−(−6) = 6
integer / entero	A whole number or its _____.	−14, 0, 6

Key idea

The sign shows the _____ of a change from a starting point.

−(−9) means the opposite of −9, which is _____.

Worked example

A water tank gains 25 liters on Monday and loses 25 liters on Tuesday. Write an integer for each day.

Zero means _____ in the water.

Monday's gain is _____. Tuesday's loss is _____.

Answer: The two changes are _____: same size, opposite effects.

You try

1. Find each value.
a. −(−11) = _____
b. opposite of the opposite of 9 _____

_____

2. A diver is at −14 meters. She moves to the opposite of that reading, then to the opposite again. Where is she now?

_____

Sum it up: The opposite of the opposite of a number is ___ because ___.

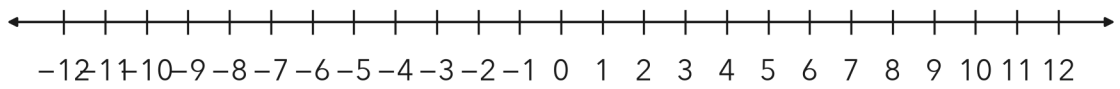
_____

Session 1 · Model, reason, and explain

Integers and Opposites · 6.NS.C.5

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A tank is at its reference level. It gains 8 L, then loses 8 L. Represent each change as an integer. Distinguish a change from the tank's actual amount.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: confuses zero change with zero quantity.. Explain what changes and what stays true.

_____

_____

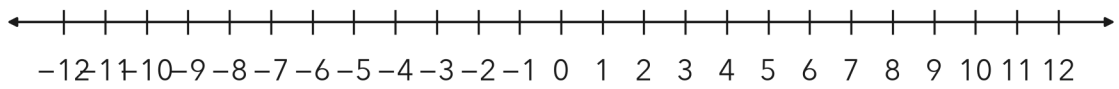
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Integers in Real Situations · 6.NS.C.5

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare a balance of −12 dollars with a change of −12 dollars. Explain the same sign in two different roles and give the opposite of −(−12).



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: treats taking the opposite twice as making the value positive in every case.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?